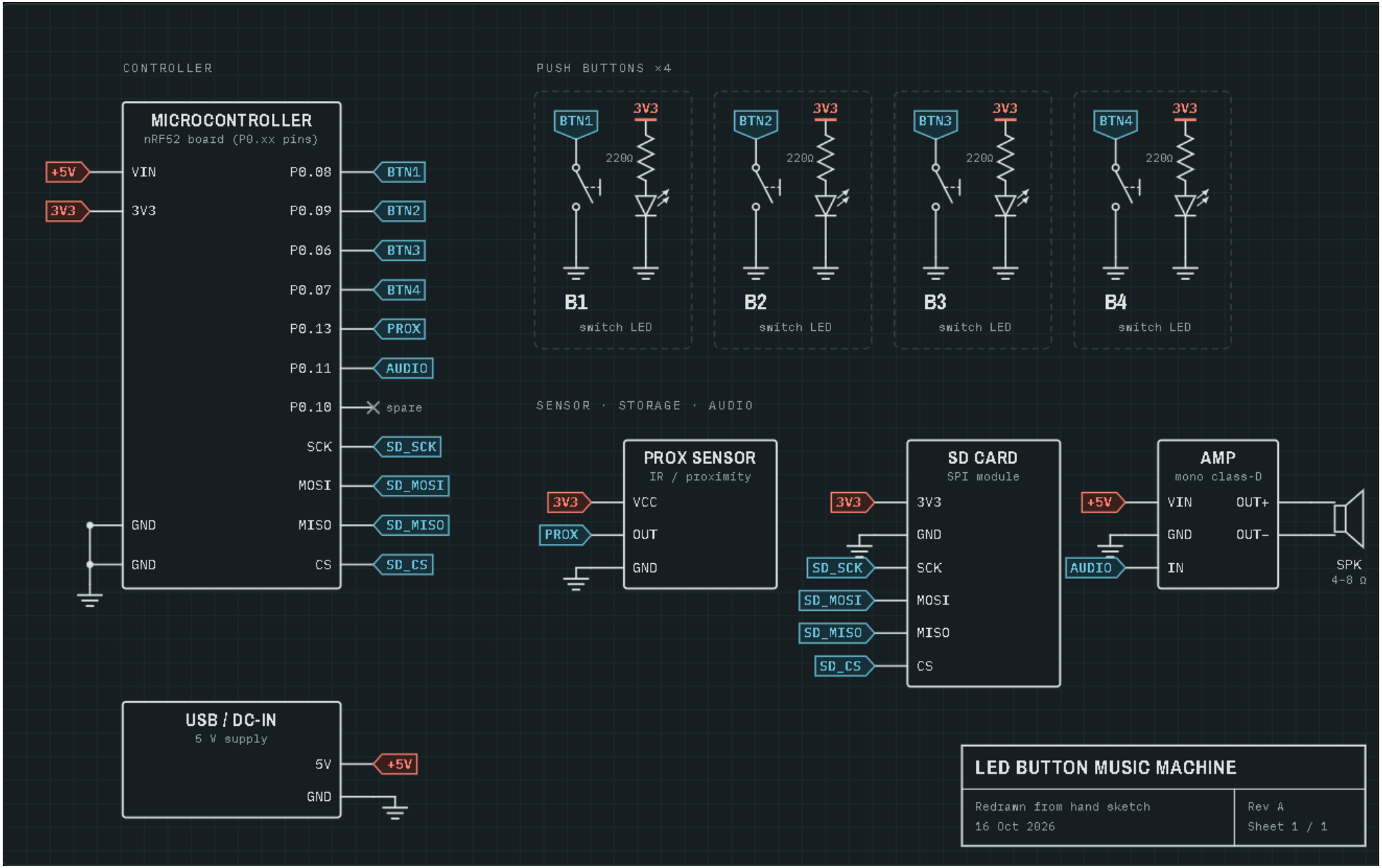

CONTROLLER
MICROCONTROLLER
nRF52 board (P0.xx pins)
+5V
3V3
VIN
3V3
GND
GND
P0.08
P0.09
P0.06
P0.07
P0.13
P0.11
P0.10
SCK
MOSI
MISO
CS
BTN1
BTN2
BTN3
BTN4
PROX
AUDIO
spare
SD_SCK
SD_MOSI
SD_MISO
SD_CS
PUSH BUTTONS ×4
BTN1
3V3
220Ω
B1
switch LED
BTN2
3V3
220Ω
B2
switch LED
BTN3
3V3
220Ω
B3
switch LED
BTN4
3V3
220Ω
B4
switch LED
SENSOR · STORAGE · AUDIO
PROX SENSOR
IR / proximity
3V3
PROX
VCC
OUT
GND
SD CARD
SPI module
3V3
SD_SCK
SD_MOSI
SD_MISO
SD_CS
3V3
GND
SCK
MOSI
MISO
CS
AMP
mono class-D
+5V
AUDIO
VIN
GND
IN
OUT+
OUT−
SPK
4–8 Ω
USB / DC-IN
5 V supply
5V
GND
+5V
LED BUTTON MUSIC MACHINE
Redrawn from hand sketch
16 Oct 2026
Rev A
Sheet 1 / 1